SECTION A (25 marks)
This section consists of **3** questions. Answer **all** questions.

1. Evaluate the following limits, if exists:

   (a) $\lim\limits_{x \to -\infty} \dfrac{3x}{\sqrt{x^2 - 36}}$. [4 *marks*]

   (b) $\lim\limits_{x \to 3} \dfrac{\sqrt{x^2 + 7} - 4}{x^2 - 3x}$. [5 *marks*]

2. (a) Find the derivative of $f(x) = \sqrt{8x - 3}$ by using the first principle of differentiation. [4 marks]

   (b) Find $\dfrac{dy}{dx}$ for the following:

   (i) $y = \sin(\ln 3x)$ [3 *marks*]

   (ii) $y = \dfrac{x}{x^3 + 1} + e^{2x} \tan x.$ [4 *marks*]

3. Determine the relative maximum and the relative minimum points of $f(x) = x^3 + 3x^2 - 24x + 20$ by using the second derivative test. [5 *marks*]

SECTION B (75 marks)
This section consists of **7** questions. Answer **all** questions.

1. Given $z_1 = 3+4i$ and $z_2 = 5-2i$. Find z_3 in $a+bi$ form if $\frac{1}{z_3} = \frac{1}{z_1} + \frac{1}{z_1 z_2}$.

   Hence, express $\overline{z_3}$ in polar form.

   [9 marks]

2. (a) Solve the equality $|3x+1| = |2x-7|$.

   [4 marks]

   (b) Obtain solution set for the following inequalities

   (i) $|2x+7| + 1 \geq 6x$. [4 marks]

   (ii) $\left|\frac{2x+1}{x-3}\right| \geq 2$. [4 marks]

3. (a) A function $f(x)$ is defined as

$$f(x) = \frac{3 - \ln(4-x)}{2}, \quad x \in \Re \quad \text{and} \quad x < 4.$$

   (i) Determine the inverse function, $f^{-1}(x)$.

   [3 marks]

   (ii) Determine whether $f^{-1}(x)$ is one to one.

   [3 marks]

   (b) Given the function $g(x) = e^{x+1}$. Find the function $h(x)$ if $(h \circ g)(x) = x+1$.

   [6 marks]

4. A function f is defined as

$$f(x) = -(x-2)^2 + 1 \quad , \quad x \geq 2$$

   Determine the inverse function of f.
   Hence, sketch the graph of f and f^{-1} on the same axes.

   [6 *marks*]

5. (a) Find the vertical and horizontal asymptotes of

   .
$$f(x) = \frac{2x^2 + 1}{4x^2 - 25}$$

   [6 *marks*]

(b) Given

$$f(x)=\begin{cases}\dfrac{x^2-16}{x^2-4x}, & x\neq 4\\ p & x=4\end{cases}$$

where p is a constant.

(i) Determine $\lim_{x\to 4} f(x)$ if exists . [3 *marks*]

(ii) Find the value of p if $f(x)$ is continuous at $x = 4$. [2 *marks*]

6. (a) The parametric equations of a curve is given by $x=-9\sin^3 t$ and $y=3-3\cos^3 t$.

(i) Show that $\dfrac{dy}{dx}=-\dfrac{1}{3}\cot t$. Hence, solve $\dfrac{dy}{dx}=0$ for $0\le t\le \pi$. [8 *marks*]

(ii) Show that $\dfrac{d^2y}{dx^2}=-\dfrac{1}{81\sin^4 t\cdot\cos t}$. [4 *marks*]

(b) Given $e^{-2y}+2xy+\ln(1-2x)=2$. Calculate the value of $\frac{dy}{dx}$ when $x = 2$ and y=0.
[4 *marks*]

7. (a) Given a water tank which has the shape of an inverted circular cone water tank with a base radius of 3 meters and a height of 5 meters. If water is being pumped into the tank at the rate of 2 m^3/minute, find the rate of change in water level rising when the water is 3 meters deep.
[5 *marks*]

(b) Air is pumped into a spherical balloon at the rate of 12π cm^3 s^{-1}.
Find the rate of increase of the surface area when its radius is 6 cm.
[4 marks]

END OF QUESTION

Final Answer:

SECTION A

1. (a) -3 (b) $\frac{1}{4}$

2. (a) $\frac{4}{\sqrt{8x-3}}$ (b)(i) $\frac{cos(ln\,x)}{x}$ (b)(ii) $\frac{1-2x^3}{\left(x^3+1\right)^2}+e^{2x}\left(2\tan x+\sec^2 x\right)$

3. Relative Minimum (2,-8) , Relative Maximum (-4, 100)

SECTION B

1. $z_3=\frac{11}{4}+\frac{13}{4}i$; $\overline{z_3}=4.26\left(\cos\left(-0.8685\right)+i\sin\left(-0.8685\right)\right)$

2. (a) $x=\frac{6}{5}$, x = -8

   (b)(i) { x: $x \le 2$} (ii) {x: $x \ge \frac{5}{4}$, $x \neq 3$ }

3. $(a)(i)$ $f^{-1}\left(x\right)=4-e^{3-2x}$ (ii) $f^{-1}\left(x\right)$ *is one to one*

   (b) $h\left(x\right)=\ln x$

4. $f^{-1}(x)=2+\sqrt{1-x}$, $x \le 1$, D.I.Y

5. (a) V.A: $x=\frac{5}{2}$, x = -$\frac{5}{2}$, H.A: $y=\frac{1}{2}$ (b) (i) 2 (ii) 2

6. (a) (i) $\frac{\pi}{2}$ (b) $\frac{dy}{dx}=-\frac{1}{3}$

7. (a) $\frac{50}{81\pi}$ (b) 4π cm^2 s^{-1}